Question Parser Training

- Pre-training with 100-300 question-program pairs for 20k iterations.
- Post-training with 9K-700k questions-solution pairs for 2M iterations with early stopping.
- Reward: 1 if program execution leads to correct answer; 0 otherwise.
- REINFORCE as the training algorithm.

REINFORCE

Optimizing the expected return $J(\theta)$:

$$\theta^\star = \arg\max_\theta J(\theta) = \arg\max_\theta E_{\tau\sim p_\theta(\tau)}\left[\sum_t r(\mathbf{s}_t, \mathbf{a}_t)\right] \tag{1}$$

$$J(\theta) = E_{\tau\sim p_\theta(\tau)}[r(\tau)] = E_{\tau\sim p_\theta(\tau)}[\sum_{t=1}^{T} r(\mathbf{s}_t, \mathbf{a}_t)] = \int p_\theta(\tau) r(\tau) d\tau \tag{2}$$

$$\nabla_\theta J(\theta) = \int \nabla_\theta p_\theta(\tau) r(\tau) d\tau = \int \frac{p_\theta(\tau)}{p_\theta(\tau)} \nabla_\theta p_\theta(\tau) r(\tau) d\tau \tag{3}$$

$$\nabla_\theta J(\theta) = \int p_\theta(\tau) \frac{\nabla_\theta p_\theta(\tau)}{p_\theta(\tau)} r(\tau) d\tau = \int p_\theta(\tau) \nabla_\theta \log p_\theta(\tau) r(\tau) d\tau \tag{4}$$

$$\nabla_\theta J(\theta) = E_{\tau\sim p_\theta(\tau)}\left[\nabla_\theta \log p_\theta(\tau) r(\tau)\right] \tag{5}$$

REINFORCE (cont.)

$$p_\theta(\tau) = p_\theta(\mathbf{s}_1, \mathbf{a}_1, ..., \mathbf{s}_T, \mathbf{a}_T) = p(\mathbf{s}_1) \prod_{t=1}^{T} \pi_\theta(\mathbf{a}_t|\mathbf{s}_t) p(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t) \tag{6}$$

we take log of both sides.

$$\log p_\theta(\tau) = \log p(\mathbf{s}_1) + \sum_{t=1}^{T} \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) + \log p(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t) \tag{7}$$

REINFORCE (cont.)

$$\theta^\star = \arg\max_\theta J(\theta) \tag{8}$$

$$J(\theta) = E_{\tau\sim p_\theta(\tau)}[r(\tau)] \tag{9}$$

$$\nabla_\theta J(\theta) = E_{\tau\sim p_\theta(\tau)}\left[\nabla_\theta \log p_\theta(\tau) r(\tau)\right] \tag{10}$$

$$\nabla_\theta J(\theta) = E_{\tau\sim p_\theta(\tau)}\left[\nabla_\theta \left[\log p(s_1) + \sum_{t=1}^{T} \log \pi_\theta(a_t|s_t) + \log p(s_{t+1}|s_t, a_t)\right] r(\tau)\right] \tag{11}$$

$$\nabla_\theta J(\theta) = E_{\tau\sim p_\theta(\tau)}\left[\left(\sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)\right)\left(\sum_{t=1}^{T} r(\mathbf{s}_t, \mathbf{a}_t)\right)\right] \tag{12}$$

REINFORCE (cont.)

$$\nabla_\theta J(\theta) \approx \frac{1}{N}\sum_{i=1}^{N}\left(\sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_{i,t} \mid \mathbf{s}_{i,t})\right)\left(\sum_{t=1}^{T} r(\mathbf{s}_{i,t}, \mathbf{a}_{i,t})\right) \tag{13}$$

$$\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta) \tag{14}$$

REINFORCE (cont.)

REINFORCE algorithm:

1. Sample $\{\tau^i\}$ from $\pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)$ (run the policy).
2. $\nabla_\theta J(\theta) \approx \sum_i \left(\sum_t \nabla_\theta \log \pi_\theta(\mathbf{a}_t^i \mid \mathbf{s}_t^i)\right)\left(\sum_t r(\mathbf{s}_t^i, \mathbf{a}_t^i)\right)$.
3. $\theta \leftarrow \theta + \alpha \nabla_\theta J(\theta)$.

REINFORCE (cont.)

- Generate samples (i.e., run the policy).
- Fit a model to estimate return.
- Improve the policy.